

Sustainability Practices Notes

5.1 ZERO WASTE

- **Definition:** Zero waste is a set of principles focused on waste prevention, encouraging the redesign of resource life cycles so that all products are reused.

5.1.1 Goal

- Reuse materials until the optimum level of consumption is reached.
- Provide guidelines for continually working towards eliminating waste.
- Avoid sending trash to landfills, incinerators, or the ocean.

5.1.2 Concept

- Conservation of all resources through responsible production, consumption, reuse, and recovery of products, packaging, and materials.
- Avoid burning and discharges to land, water, or air that threaten the environment or human health.
- **Examples of Non-Zero Waste:**
 - One-way recyclable glass bottles.
 - One-way milk bags.
 - One-way aseptic cartons.
 - One-way table-top paper board cartons.

5.1.3 Principles of Zero Waste

1. **Refuse:** Refuse what you don't need to prevent unwanted items from entering your home.
2. **Reduce:** Reduce what you use to minimize waste generation.
3. **Reuse:** Reuse items whenever possible.
4. **Recycle:** Recycle what you can't refuse or reduce.
5. **Regulate:** Compost food scraps, paper pieces, and wooden/bamboo toothbrushes to return nutrients to the earth.

5.1.4 Steps to Achieve Zero Waste

1. Identify high waste areas in your lifestyle.
2. Apply the principle of zero waste where waste cannot be removed or reduced.
3. Substitute single-use plastics with eco-friendly, zero-waste options.
4. Buy zero-waste or eco-friendly products.

5. Support eco-friendly businesses.
6. Compost kitchen waste.
7. Reuse, upcycle, and repurpose items.

5.1.5 Advantages of Zero Waste

- Reduces climate impact.
- Conserves resources and minimizes pollution.
- Promotes social equity and builds community.
- Supports a local circular economy and creates jobs.

Disadvantages of Zero Waste

- Difficult to store solid wastes.
- Can be more expensive.
- Time-consuming.
- Can cause anxiety.
- Can be misleading.
- Difficult for large households.
- Zero waste products can be hard to find.

5.2 R CONCEPT (REDUCE, REUSE, RECYCLE)

Definition

The principle of reducing waste, reusing, and recycling resources and products.

- **Reduce:**
 - Using things with care to reduce waste generation.
 - Reducing raw material usage to decrease waste generation.
- **Reuse:**
 - Repeated use of items or parts of items with usable aspects.
 - Examples:
 - Refillable containers.
 - Rubber rings from discarded cycle tubes.
- **Recycle:**
 - Using waste as a resource.
 - Reprocessing discarded materials into new, useful products.
 - Examples:
 - Melting old aluminum cans and glass bottles into new ones.
 - Preparing cellulose insulation from paper.
 - Preparing fuel pellets from kitchen waste.
 - Preparing automobiles and construction materials from steel cans.

- Saves money, energy, raw materials, and reduces pollution.

5.2.1 Concept of 3R

- Refers to reduce, reuse, and recycle, particularly in production and consumption.
- Aims for an increase in the ratio of recyclable materials.
- Promotes further reusing of raw materials and manufacturing wastes.
- Overall reduction in resources and energy used.

5.2.2 Principle

- 3R is the order of priority of actions to reduce waste generation and improve overall waste management.

5.2.3 Importance of 3 Rs

- Reducing waste at the source is the most effective way to reduce garbage.
- Saves water and energy.
- Reusing avoids creating waste.
- Well-run recycling programs cost less than waste collection and landfilling.
- Recycling helps families save money due to less disposal costs.
- Reduces air and water pollution compared to manufacturing with new materials.
- Less material is sent to landfills, preserving them for the future.
- Proper disposal and recycling prevents water and soil contamination.

5.2.4 Advantages of 3 Rs

- Reduce greenhouse gas emissions.
- Saves energy.
- Helps sustain the environment for future generations.
- Reduces the amount of waste sent to landfills and incinerators.
- Save money.
- Prevent pollution.

Disadvantages of 3 Rs

- High upfront capital cost.
- Recycling sites are often unhygienic, unsafe, and unsightly.
- Products from recycled waste may not be durable.
- Recycling may not always be inexpensive.
- Can be more energy consumption and pollution.
- Generates pollutants.
- High processing cost.
- Low quality of resultant product.

5.3 CIRCULAR ECONOMY

Definition

A new production and consumption model that ensures sustainable growth over time. It reduces the consumption of raw materials and recovers wastes by recycling or giving it a second life as a new product.

5.3.1 Aim

Make the most of available material resources by applying the reduce, reuse, and recycle principles, extending the life cycle of products and establishing a more efficient and sustainable production model.

5.3.2 Benefits of Circular Economy

- Protects the environment.
- Benefits the local economy.
- Drives employment growth.
- Promotes resource independence.

5.3.3 Necessary Steps (7Rs) to Achieve a Circular Economy

1. **Redesign:** Consume fewer raw materials, extend life cycles, and generate less waste.
2. **Reduce:** Reduce consumption, waste generation, and the use of raw materials.
3. **Reuse:** Extend the life cycle of products by reusing them.
4. **Repair:** Avoid using new raw materials, save energy, and prevent environmental waste by repairing products.
5. **Renovate:** Update old objects for reuse.
6. **Recycle:** Use waste products as raw materials to manufacture new products.
7. **Recover:** Use discarded products for new purposes.

Example

- Manufacturers designing products to be reusable.
- Designing electrical devices for easier repair and reusing products and raw materials as much as possible.

5.4 ISO 14000 SERIES

Definition

An International Organization for Standardization composed of representatives from various national standard organizations, providing standards and guidelines for businesses and purposes and publishes technical reports.

- ISO 14000 Series: Family of standards related to environmental management to help organizations:
 - Minimize negative environmental impacts.

- Comply with applicable laws and regulations.
- Continually improve environmental performance.

5.4.1 Objective of ISO14000 Series

Promote effective environmental management systems in organizations.

5.4.2 List of ISO 14000 Series Standards

It includes a catalog of over 50 Environmental management and performance related standards. But some important ISO 14000 series standards are listed here.

Standard	Title	Applications
ISO14001	Environmental management system	Requirements with guidance for use.
ISO14004	Environmental management system	General guidelines on implementation.
ISO14005	Environmental management system	Guidelines for flexible approach to implementation.
ISO14015	Environmental management	Environmental assessment of sites.
ISO14020 to 14025	Environmental management	Environmental labels and declarations.
ISO14030	Green bonds	Environmental performance of nominated projects and assets.
ISO14031	Environmental management	Environmental performance evaluation & Guidelines.
ISO14040 to 14049	Environmental management	Discusses pre-production planning.
ISO14050	Environmental management	Vocabulary, terms and definitions.
ISO14062	Environmental management	Product design and development.
ISO14063	Environmental management	Guidelines and examples.
ISO14064	Environmental management	Reducing green house gas emissions.
ISO14090	Environmental management	Adaptation of climate change.

5.4.3 Core Elements of ISO 14000

1. Environmental policy.

2. Planning.
3. Implementation and operation.
4. Checking and corrective action.
5. Management review.
6. Continuous improvement.

5.4.4 Advantages of ISO14000

- Identifies risks and opportunities.
- Prevents problems from reoccurring.
- Boosts marketing and sales efforts.
- Improves employee performance.
- Improves control over the business.
- Lowers costs like energy bills, tax, and insurance bills.
- Helps to reduce waste.
- Helps to minimize the carbon footprint of a company.
- It is recognized internationally.
- It gives immediate notice about the environmental performance of a company.

Disadvantages of ISO14000

1. Extremely costly to implement if not done properly.
2. Requires a lot of administrative work.
3. No guarantee of improvement in environmental performance.
4. Organizations face challenges while implementing these standards.

5.5 MATERIAL LIFE CYCLE ASSESSMENT (LCA)

Definition

A process of evaluating the environmental effects of a material over its entire life cycle, increasing resource use efficiency and decreasing liabilities. Commonly referred to as a cradle-to-grave analysis.

5.5.1 Stages of a Life Cycle Assessment

1. Raw materials extraction and processing.
2. Manufacturing.
3. Transportation.
4. Distribution.
5. Usage and retail.
6. Waste disposal (end of life).

5.5.2 Benefits of LCA

1. Supports sustainable development.
2. Allows decision-makers to compare products and select the one with the lowest environmental impact.
3. Modelling tool to assess environmental impacts of a product throughout its lifespan.
4. Provides a holistic view of environmental impacts, avoiding optimization of one indicator without considering effects on others.
5. Identifies hotspots in the environmental impact.
6. Based on internationally accepted standards.

5.5.3 Disadvantages of LCA

1. Assesses the real world with a simplified model.
2. Assumptions, scenarios, and scope vary, leading to different results.
3. Variations in approaches and results may be confusing for non-experts.
4. Requires a large amount of data.
5. Poor data collection leads to inconclusive results.
6. Difficult to communicate results.

5.6 ENVIRONMENTAL IMPACT ASSESSMENT (EIA)

Definition

A formal process of predicting the environmental consequences of a development project, used to identify environmental, social, and economic impacts before decision-making.

5.6.1 Purpose of EIA

Determine the potential environmental, social, and health effects of proposed development projects.

5.6.2 Objectives of EIA

1. Identify the main issues and problems of involved parties.
2. Identify who the parties are.
3. Identify the problems of the parties.
4. Identify why the problems arise.

5.6.3 Benefits of EIA

1. Cost and time of the project is reduced.
2. Performance of the project is improved.
3. Waste treatment and cleaning expenses are minimized.
4. Usages of resources are decreased.
5. Biodiversity is maintained.
6. Human health is improved.
7. It helps in preventing natural calamities like earthquake, cyclone, etc.,

5.6.4 Process of EIA (Key Elements)

1. **Scoping:** Identify key issues of concern early in the planning process and aid site selection.
2. **Screening:** Decide whether an EIA is required based on collected information.
3. **Identifying and Evaluating Alternatives:** Consider alternative sites and techniques and their potential impacts.
4. **Mitigating Measures:** Review actions to prevent or minimize adverse effects of a project.
5. **Environmental Statements:** Report the findings of the EIA.

5.7 SUSTAINABLE HABITAT

Definition

An ecosystem that produces food and shelter for people and other organisms without resource depletion.

5.7.1 Features of a Sustainable Habitat

- Proper waste management.
- Affordable housing.
- Wastewater treatment and recycling.
- Green transportation using green fuel like biodiesel.

5.7.2 Objectives of National Mission on Sustainable Habitat

1. Reduce energy demand by promoting alternative technologies and energy conservation practices.
2. Improve urban planning with disaster management, reduced private transport, and increased public transport.
3. Encourage community involvement and stakeholder participation.
4. Conserve natural resources such as clean air, water, flora, and fauna.
5. Facilitate the growth of small and medium cities.
6. Engineers and architects should not consider any element as a waste product.

How to Maintain a Sustainable Habitat

- Promote energy efficiency.
- Promote the use of eco-friendly fuels.
- Better manage municipal solid waste.
- Promote public transport.

5.8 GREEN BUILDINGS

Definition

An efficient method of construction that produces healthier buildings with less environmental and climate impact that requires less cost to maintain. Green buildings preserve precious natural resources and improve the quality of life.

5.8.1 Criteria for Green Buildings

1. Build on previously developed land.
2. Build near existing infrastructure.
3. Smaller building site to reduce environmental footprint.
4. Sustainably landscaped sites that don't suffer from soil erosion.
5. Water reduction through low-flow toilets and grey water systems.
6. Clean energy use like geothermal, solar, wind energies.
7. Reduce material usage and use natural, renewable sources.
8. Use low emitting materials to improve human health and protect the environment.

5.8.2 Features of Green Buildings

- Efficient use of energy, water, and other resources.
- Use of renewable energy.
- Pollution and waste reduction through reuse and recycling.
- Good indoor environmental air quality.
- Use of non-toxic, ethical, and sustainable materials.
- Design enables adaptation to a changing environment.
- Consideration of the quality of life of occupants.
- Environmentally conscious construction and operation.

5.8.3 Principles of Green Building

1. Livable communities.
2. Energy efficiency.
3. Indoor air quality.
4. Resource conservation.
5. Water conservation.

5.8.4 Components of Green Building

1. Aluminum weather-resistant insulated access panel.
2. Energy-efficient windows.
3. Green roof.
4. Solar power.
5. Water conservation.
6. Recycling.
7. Landscaping.

5.8.5 Advantages of Green Buildings

- Energy efficient.
- Higher fraction of eco-friendly materials.
- Water-efficient devices.
- Reduction in waste.
- Less air pollution.
- Reduction in greenhouse gas emissions.
- Protection of natural resources.
- Improved indoor air quality.
- Use of recycled metal and other construction materials.
- Renewable energies emphasis.
- Daylighting is utilized as best as possible.
- Use of renewable plant materials.
- Higher market value.
- Rainwater collection and use of compost bins.
- Overall health improvements.

Disadvantages of Green Buildings

- High initial costs.
- Energy supply may depend on weather conditions.
- Technology problems are more.
- Maintenance may be difficult.
- Indoor air temperature may greatly vary over time.
- Experienced green construction workers may be rare.
- Green construction is not suitable for all locations.
- Availability of green construction materials.
- Funding problems.

5.9 GREEN MATERIALS

Definition

Eco-friendly building construction materials that have a low impact on the environment, widely used in various industrial applications due to non-toxic, organic, and recyclable properties.

- Examples: Naturally occurring materials like wood, ceramics, glass, clay, sand, stone.

5.9.1 Criteria for Green Materials

1. Local availability.
2. Embodied energy.
3. Percentage of recycled/waste materials used.

4. Rapidly renewable materials.
5. Contribution to energy efficiency.
6. Recyclability.
7. Durability.
8. Environmental impact.

5.9.2 Characteristics of Green Materials

1. Energy efficient, using less energy for the same task.
2. Lowers energy costs and lessens pollution.
3. Mostly renewable and can be regenerated.
4. Recyclable or made from recycled material, saving energy and reducing waste.
5. Non-toxic, emitting no odors, irritants, or hazardous compounds.
6. Durable and preserve resources and energy.
7. Cost-effective.
8. Can be locally sourced to reduce transport costs.

5.9.3 Important Green Building Materials

Green building is construction that primarily uses natural materials and renewable resources.

1. Stone: durable and low-maintenance.
2. Cob: (mud mixture of natural ingredients like soil, sand, straw and lime). cheap and energy efficient.
3. Bamboo: durable and light weight.
4. Cork: (Cork canes from oak trees). A very good thermal insulator and mold resistant.
5. Adobe brick: (brick made of clay and straw). Natural noise protection and posses unique design (can be easily cut and transformed).
6. Straw bale: Easily renewable and cheap.
7. Cord wood: Affordable, thermal efficiency.
8. Earth bags (or) sand bags: Locally sourced and provide natural insulation.
9. Mycelium (or) mushroom roots: Strong and light weight.

5.9.4 Examples of Green Materials

- Bamboo floorings.
- LED lightings.
- Reclaimed wood.
- Energy-efficient appliances.
- High-efficiency glass windows.
- Solar panels.
- Recycled steel.
- Cork.

- Precast concrete slabs.
- Low VOC paint.

5.10 ENERGY EFFICIENCY

Definition

The use of less energy to perform the same task or produce the same result.

5.10.1 Methods of Achieving Energy Efficiency

- Alternative waste treatment.
- Avoided emissions from diverting legacy waste from landfill for process engineered fuel manufacture.
- Avoided emissions from diverting legacy waste from landfill through a composting alternative waste technology.
- Capture and combustion of landfill gas.

5.10.2 Calculation of Energy Efficiency

$$\text{Energy Efficiency} = \frac{\text{energy output}}{\text{energy input}} \times 100\%$$

$$\eta = \frac{W_{out}}{W_{in}} \times 100\%$$

5.10.3 Advantages of Energy Efficiency

1. Saves money.
2. Lowers greenhouse gas emissions and other pollutants.
3. Decreases water use.
4. Lowers utility bills, creates jobs, and helps stabilize electricity prices.
5. Provides long-term benefits by lowering overall electricity demand.
6. Environmentally friendly as it does not emit harmful carbon dioxide into the atmosphere.

Example

Energy-efficient LED light bulbs produce the same amount of light as incandescent light bulbs using 75 to 80% less electricity.

5.10.4 Disadvantages of Energy Efficiency

1. High cost of enforcing energy-efficient construction.
2. Building materials are not always available.
3. Slight unfavorable effects on human health.
4. Indoor air is 3 to 7 times more polluted than outdoor air.

5.11 SUSTAINABLE TRANSPORT

Definition

Green transportation that has a low impact on the environment.

Examples:

1. walking
2. cycling
3. transit
4. carpooling
5. car sharing
6. green vehicles

5.11.1 Importance of Sustainable Transport

1. Contributes to a reduction in damaging CO₂ emission and improved air quality in cities.
2. Reduces the negative impacts on the environment.

5.11.2 Key Elements of Sustainable Transport

1. **Fuel economy:** The better fuel economy gets the lower emissions go.
2. **Occupancy:** Lower the carbon intensity of a vehicle is to stick more people in the vehicle.
3. **Electrification:** Most important pathway to low carbon transport.
4. **Pedal power:** Bicycles reduces the carbon emissions.
5. **Urbanization:** Opportunity for lowering both distance travelled per person and the carbon intensity of that travel.

5.11.3 How to Promote Sustainable Transport

1. Enhancing public transportation: Promoting HSE (Health, safety and environment) policy.
2. Encouraging car pooling: Reduces the volume of CO₂, emitted per inhabitant.
3. Encouraging bicycle use: Reliable and non-polluting means of transportation.
4. Teleworking: It reduces employee travel.
5. Improving the parking experience.

5.11.4 Advantages of Sustainable Transport

- Creates jobs.
- Provides safer transportation.
- Emits less pollution.
- Promotes health.
- Saves energy.
- Saves money.
- Decreases congestion.

- Conserves land.

Disadvantages of Sustainable Transport

- Modifications to handling and transport facilities.
- The initial purchase of reusable containers.
- Additional costs of the tracking system.

5.12 SUSTAINABLE ENERGY

Definition

Energy that meets the needs of the present without compromising the ability of future generations to meet their own needs.

5.12.1 Sources of Sustainable Energy

- Wind energy.
- Solar energy.
- Ocean energy.
- Hydro power.
- Geothermal energy.

5.12.2 Advantages of Sustainable Energy

1. Improves public health.
2. Creates local jobs.
3. Decrease your carbon footprint.
4. Cost saving.
5. Energy security.

Disadvantages of Sustainable Energy

1. Sustainable energy sources are not available continuously.
2. The efficiency of sustainable energy technologies is low.
3. The initial cost of sustainable energy is high.
4. Sustainable energy sites require a lot of space.
5. Sustainable energy devices need recycling.

5.13 NON-CONVENTIONAL (RENEWABLE) ENERGY SOURCES

Definition

Natural resources that can be regenerated continuously and are inexhaustible.

- **Examples:** Wood, solar energy, wind energy, hydropower, tidal energy, etc.

Merits of Non-Conventional Energy Resources

1. Unlimited supply.
2. Provides energy security.
3. Fits into sustainable development concept.
4. Reliable and modular in size.
5. Decentralized energy production.

5.13.1 Solar Energy

Energy directly from the sun.

- **Methods of Harvesting Solar Energy**

- Solar cells (or) photovoltaic cells (or) PV cells

Uses Used in calculators, electronic watches, street lights, water pumps to run radios and TVs.

Solar Battery

When a large number of solar cells are connected in series it form a solar battery.

Used in remote areas where conventional electricity supply is a problem.

Solar heat collectors

Solar heat collectors consist of natural materials like stones, bricks (or) materials like glass, which can absorb heat during the day time and release it slowly at night.

It is generally used in cold places, where houses are kept in hot condition using solar heat collectors.

Solar water heater

It consists of an insulated box inside of which is painted with black paint. It is also provided with a glass lid to receive and store solar heat. Inside the box it has black painted copper coil, through which cold water is allowed to flow in, which gets heated up and flows out into a storage tank. From the storage tank water is then supplied through pipes.

5.13.2 Wind energy

Moving air.

Wind mills

The strike of blowing wind on the blades of the wind mill makes it rotating continuously. The rotational motion of the blade drives a number of machines like water pump, flour mills and electric generators.

Wind farms

When a large number of wind mills are installed and joined together in a definite pattern it forms a wind farm.

- The minimum speed required for satisfactory working of a wind generator is 15 km/hr .

Advantages of Wind energy

1. It does not cause any air pollution.

2. It is very cheap.

5.13.3 Ocean energy

Ocean can also be used for generating energy in the following ways.

(1) Tidal energy (or) Tidal power

Ocean tides, produced by gravitational forces of sun and moon, contain enormous amount of energy can be harnessed by constructing a tidal barrage.

During high tide, the sea-water is allowed to flow into the reservoir of the barrage and rotates the turbine, which in turn produces electricity by rotating the generators. During low tide, when the sea level is low, the sea water stored in the barrage reservoir is allowed to flow into the sea and again rotates the turbine.

(2) Ocean thermal energy (OTE)

Utilized to generate electricity. The temperature difference should be of 20°C (or) more is required between surface water and deeper water.

Process The warm surface water of ocean is used to boil a low boiling liquid like ammonia. The high vapour pressure of the liquid, formed by boiling, is then used to turn the turbine of the generator and generates electricity. The cold water from the deeper ocean is pumped to cool and condense the vapour into liquid.

(3) Geo-thermal energy

Temperature of the earth increases at a rate of $20 \text{ to } 75^{\circ}\text{C}$ per km, when we move down the earth surface.

High temperature and high pressure steam fields exist below the earth's surface in many places. Natural geysers In some places, the hot water (or) steam comes out of the ground through cracks naturally in the form of natural geysers. Artificial geysers In some places, we can artificially drill a hole up to the hot region and by sending a pipe in it, we can make the hot water (or) steam to rush out through the pipe with very high pressure.

5.13.4 Biomass energy

Biomass is the organic matter, produced by plants (or) animals, used as sources of energy. Energy example wood, crop residues, seeds, cattle dung, sewage, agricultural wastes, etc.,

Biogas

Biogas is a mixture of gases such as methane, carbon dioxide, hydrogen sulphide, etc., It contains about 65% of methane gas as a major constituent.

Biofuels

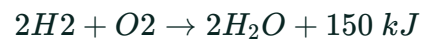
Biofuels are the fuels, obtained by the fermentation of biomass.

Example Ethanol, methanol.

- Ethanol: produced from the sugarcane
- Methanol: Methanol can be easily obtained from ethanol (or) sugar-containing plants.
- Gasohol: Gasohol is a mixture of ethanol + gasoline

Hydrogen Fuel

Hydrogen can be produced by thermal dissociation (or) photolysis (or) electrolysis of water it possesses high calorific value.



Disadvantages of hydrogen fuel

1. Hydrogen is highly inflammable and explosive in nature.
2. Safe handling is required.
3. It is difficult to store and transport.

5.14 ENERGY CYCLES

Energy cycle, is the interactions between energy sources within the Earth's environment. These interactions are very complex and even small changes in them can lead to significant changes in long-term climate behavior.

Important energy cycles

1. Carbon cycle.
2. Nitrogen cycle.
3. Phosphorus cycle.

5.14.1 Carbon cycle

Carbon cycle is the movement of carbon (or) carbon compounds continuously from the atmosphere to the earth and then back into the atmosphere.

Carbon in the atmosphere is present in the form of carbon dioxide. Carbon enters the atmosphere through natural process such as respiration and industrial applications such as burning of fossil fuels

Sources of CO₂ in atmosphere\ During respiration, plants and animals liberates CO₂ in the atmosphere.\Combustion of fuels also release CO₂.\Volcanic eruptions also release CO₂.

Various steps involved in carbon cycle

1. Carbon present in the atmosphere is absorbed by plants by the processes photosynthesis
2. These plants are then consumed by animals.
3. These animals and plants eventually die and decomposers eat the dead organism and return the carbon from their body back into the atmosphere
4. Some of the carbon that is not released back into the atmosphere eventually become fossil fuels.
5. These fossil fuels are then used for man-made activities, which pump more carbon back into the atmosphere.

Importance of carbon cycle

1. It plays a vital role in balancing the energy
2. Carbon cycle is an important aspect of the survival of all life on earth
3. Carbon is the building block of life and forms bonds with other elements necessary for life

5.15 CARBON EMISSION AND SEQUESTRATION

5.15.1 Carbon emission

Carbon emission is the release of green house gases and their precursors into the atmosphere over a specified area and period of time.

Types of carbon emissions

Carbon (Green house gas) emissions are classified into two scopes.

- Scope 1 emissions (or) Direct emissions\stationary combustion\mobile combustion\Fugitive emissions\Process emissions
- Scope 2 emissions (or) Indirect emissions

Sources of carbon emissions
[0.2cm]1) Natural sources of CO₂ emission\decomposition of matter\ocean release\respiration\Most animals, which exhale CO₂ as a \waste product\\Carbonate rocks.
[0.2cm]2) Human sources of CO₂ emission\Burning of fossil fuels like coal, natural gas and oil\\Deforestation\\Industrial activities like cement manufacture, oil refineries and leather industries.\Transportation sector generates largest amount of CO₂ in the atmosphere.

Harmful effects of carbon emissions

1. Carbon emission, affecting the planet significantly, affects climate change. and leads to global warming
2. Reduction of carbon emission\[0.2cm] energy efficiency\ fuel switching\combined heat and power\use of renewable energy\more efficient use\recycling of materials\plant more trees\reduce air travel\driving more efficient.\[0.2cm] 5.15.2 Carbon sequestration

Reduction global climate change\There are three main types of carbon sequestration.\[0.2cm]1) Biological carbon sequestration\It is the storage of CO₂ in vegetation like grassland, forests, soils and oceans.

1. Geological carbon sequestration.
2. Technological carbon sequestration. Graphene production

Advantages of carbon sequestration

1. Carbon sequestration prevents the occurrence of climate change.
2. Deep injection of CO₂ improves the extraction of fuels like oil and methane from their reserves in addition to removing excess pollutants from the air.
3. Since the gas can be easily liquefied, it can be easily transmitted through pipelines.
4. No CO₂ leaking out from the injection site.
5. It lowers carbon emission by 80% to 85% while using fossil fuels.

Disadvantages of carbon sequestration

1. Due to carbon sequestration, in power plants, 40% additional coal is consumed and hence cost of energy gets increased by 1 to 5% per kilowatt hour.

2. CO₂ from power plant emissions must be captured and liquified, which uses a lot of electrical power.
3. It can be disastrous if the injected gas leaks due to structural flaws in the geological formation.
4. The ocean can become acidic due to the large amounts of carbon dioxide being injected into it, endangering aquatic life.
5. Planting trees, with the intention of storing and absorbing carbon, requires more time for the trees to mature.
6. There is no enough available geological resources to sequester carbon.

5.16 GREEN ENGINEERING

5.16.1 Definition

Green engineering is the design, commercialization and use of processes and products that minimizes pollution, promotes sustainability and promotes human health without affecting environment.

5.16.2 Examples for green engineering

1. Biodegradable cups and straws.
2. Enhanced industrial emission filters.
3. Waste water treatment.
4. Plant-based cooling